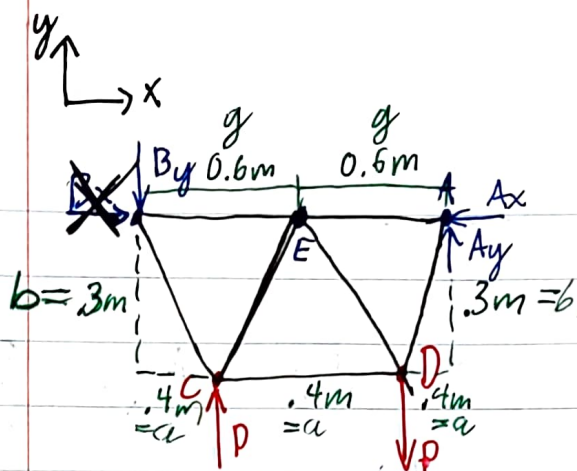




$$0.4 \times 3 = 1.2$$

$$1.2/2 = 0.6$$

External Forces

$$\Sigma F_x = 0$$

$$\Sigma F_x = \cancel{B_x} - A_x = 0$$

$$\cancel{B_x} - A_x = 0 \quad A_x = 0$$

$$\Sigma F_y = 0$$

$$\Sigma F_y = \cancel{P} - \cancel{P} + A_y + B_y = 0$$

Incorrect sign

$$A_y = -B_y$$

$$\Sigma M_B = Pa - 2Pa + A_y(3a)$$

$$= -Pa + 3aA_y$$

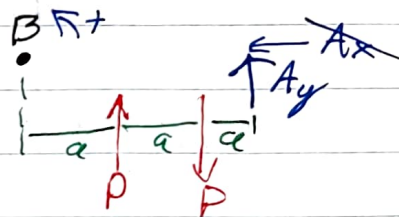
$$\Rightarrow 3aA_y = +Pa$$

$$A_y = \frac{+Pa}{3a} = +\frac{P}{3}$$

$$A_y = \cancel{\frac{P}{3}} + \frac{P}{3}$$

$$-B_y = +\frac{1}{3}P$$

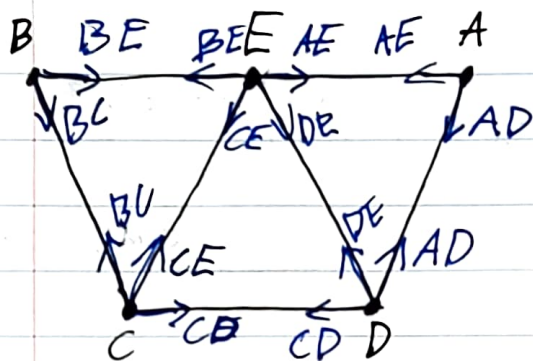
$$B_y = -\frac{1}{3}P$$



$$A_y = \frac{1}{3}P$$

$$B_y = -\frac{1}{3}P$$

wrong sign



$$A_x = 0$$

$$A_y = \frac{P}{3}$$

$$B_y = -\frac{P}{3}$$

$$a = 0.4 \text{ m}$$

$$b = 0.3 \text{ m}$$

$$d = \sqrt{0.4^2 + 0.3^2}$$

$$d = 0.5 \text{ m}$$

$$h = \sqrt{\left(\frac{a}{2}\right)^2 + b^2}$$

$$h = \sqrt{\left(\frac{0.4}{2}\right)^2 + (0.3)^2}$$

$$h = 0.360555128 \text{ m}$$

$$\checkmark AD = A_y \frac{d}{b} = \frac{P}{3} \cdot \frac{d}{b}$$

$$\checkmark AE = -A_y \frac{a}{b} = -\frac{P}{3} \cdot \frac{a}{b}$$

$$\checkmark DE = \frac{h}{b} (P - A_y) = \frac{h}{b} \left(P - \frac{P}{3}\right)$$

$$\checkmark CD = -\frac{a/2}{b} (B_y - P) + B_y \cdot \frac{d}{b} = -\frac{a/2}{b} \left(-\frac{P}{3} - P\right) + \left(-\frac{P}{3}\right) \frac{d}{b}$$

$$\checkmark CE = \frac{h}{b} (B_y - P) = \frac{h}{b} \left(-\frac{P}{3} - P\right)$$

$$\checkmark BC = -B_y \cdot \frac{d}{b} = -\left(-\frac{P}{3}\right) \frac{d}{b} = \frac{P}{3} \cdot \frac{d}{b}$$

$$\checkmark BE = B_y \cdot \frac{a}{b} = -\frac{P}{3} \cdot \frac{a}{b}$$

Setting $P = 25 \text{ kN}$ & solving numerically

External Forces

$$A_x = 0$$

$$A_y = \frac{25}{3} = 8.33 \text{ kN}$$

$$B_y = -\frac{25}{3} = -8.33 \text{ kN}$$

Internal Forces

$$\checkmark AD = \frac{P}{3} \cdot \frac{d}{b} = \frac{25}{3} \cdot \frac{0.5}{0.3} = 13.889 \text{ kN (T)}$$

$$\checkmark AE = -\frac{P}{3} \cdot \frac{a}{b} = -\frac{25}{3} \cdot \frac{0.4}{0.3} = -11.111 \text{ kN (C)}$$

$$\checkmark DE = \frac{k}{b} \left(P - \frac{P}{3} \right) = \frac{0.361}{0.3} \left(25 - \frac{25}{3} \right) = 20.031 \text{ kN (T)}$$

$$\checkmark CD = -\frac{a/2}{b} \left(-\frac{P}{3} - P \right) + \left(\frac{P}{3} \right) \frac{d}{b} = -\frac{0.4/2}{0.3} \left(-\frac{25}{3} - 25 \right) + \left(\frac{25}{3} \right) \left(\frac{0.5}{0.3} \right)$$

$$CD = 36.111 \text{ kN (T)}$$

$$\checkmark BC = \frac{P}{3} \cdot \frac{d}{b} = \frac{25}{3} \cdot \frac{0.5}{0.3} = 13.889 \text{ kN (T)}$$

~~Problem 1~~

$$\checkmark CE = \cancel{\frac{h}{b}} \cdot \frac{h}{b} \left(-\frac{P}{3} - P \right) = \frac{0.361}{0.3} \left(-\frac{25}{3} - 25 \right) = -40.061 \text{ KN (C)}$$

$$\checkmark BE = -\frac{P}{3} \cdot \frac{a}{b} = -\frac{25}{3} \cdot \frac{0.4}{0.3} = -11.111 \text{ KN (C)}$$

$$AD = 13.889 \text{ KN (T)}$$

$$AE = 11.111 \text{ KN (C)}$$

$$\underline{DE = 20.031 \text{ KN (T) Error}}$$

$$\underline{CD = 36.111 \text{ KN (T) Error}}$$

~~BC = 13.889~~

$CE = 40.061 \text{ KN (C)}$	Largest Force
------------------------------	---------------

$$BC = 13.889 \text{ KN (T)}$$

$$BE = 11.111 \text{ KN (C)}$$

Pin C

$$\Sigma F_y = P + BC \frac{b}{a} + CE \frac{b}{h} = 0$$

$$\Rightarrow 25 + (13.889) \frac{0.3}{0.5} + (-40.061) \left(\frac{0.3}{0.361} \right) = 0$$

$$\Rightarrow 25 + 8.3334 - 33.333 = 0$$

$$0 = 0 \checkmark$$

$$\Sigma F_x = CD + CE \frac{a/2}{h} - BC \frac{a}{d} = 0$$

$$\Rightarrow 36.111 + (-40.061) \left(\frac{0.4/2}{0.361} \right) - 13.889 \left(\frac{0.4}{0.5} \right)$$

$$\Rightarrow 36.111 - 22.222 - 11.111 = 0$$

$$\Rightarrow -3 \neq 0 \quad \times$$

Pin D

$$\Sigma F_y = -P + AD \frac{b}{a} + DE \frac{b}{h} = 0$$

$$= -25 + (13.889) \left(\frac{0.3}{0.5} \right) + (20.031) \left(\frac{0.3}{0.361} \right)$$

$$-0.02 \approx 0$$

$$\Sigma F_x = -CD - DE \frac{a/2}{h} + AD \frac{a}{d} = 0$$

$$\Rightarrow (36.111) - (20.031) \frac{0.4/2}{0.361} + (13.889) \left(\frac{0.4}{0.5} \right) = 0$$

$$= -38.875 \neq 0 \quad \times$$